Dataset	Paraphrased	Less Diverse	Negated	Simplified	Summarized	Synonym Replaced	Added Source Text
SQuALITY	0.023	0.033	0.681	0.029	0.018	0.034	0.052
LexAbSumm	0.004	0.001	0.560	0.002	0.007	0.013	0.030
ScholarQABench	0.010	0.017	0.542	0.013	0.006	0.028	0.019

Table 3: Average contradiction rate between perturbed and original summaries, computed using an NLI-based faithfulness check. Lower values indicate higher factual consistency.

Metric	Original	Synonym Replaced	Summarized	Simplified	Paraphrased	Negated	Less Diverse	Added Source Text
BARTScore	0.16	0.07	0.08	0.11	0.09	0.07	0.10	0.23
MiniCheck	0.84	0.83	0.84	0.85	0.85	0.40	0.85	0.78
SummaC-Conv	0.33	0.31	0.29	0.37	0.21	0.23	0.34	0.42
SummaC-ZS	0.38	0.39	0.32	0.47	0.39	0.22	0.38	0.48
AlignScore	0.52	0.48	0.42	0.56	0.38	0.38	0.51	0.58
UniEval	0.82	0.83	0.80	0.84	0.86	0.39	0.82	0.83

Table 4: Mean factuality scores for each metric and perturbation type on LexAbSumm.

Metric	Original	Synonym Replaced	Summarized	Simplified	Paraphrased	Negated	Less Diverse	Added Source Text
BARTScore	0.03	0.02	0.02	0.02	0.02	0.02	0.03	0.06
MiniCheck	0.32	0.32	0.32	0.35	0.32	0.19	0.35	0.39
SummaC-Conv	0.23	0.23	0.23	0.24	0.23	0.22	0.23	0.27
SummaC-ZS	0.18	0.19	0.19	0.22	0.18	0.13	0.19	0.23
AlignScore	0.21	0.22	0.19	0.22	0.20	0.20	0.22	0.26
UniEval	0.73	0.73	0.72	0.72	0.73	0.32	0.71	0.73

Table 5: Mean factuality scores for each metric and perturbation type on ScholarQABench.

Metric	Original	Synonym Replaced	Summarized	Simplified	Paraphrased	Negated	Less Diverse	Added Source Text
BARTScore	0.03	0.02	0.02	0.03	0.03	0.02	0.03	0.07
MiniCheck	0.56	0.55	0.53	0.55	0.56	0.30	0.55	0.57
SummaC-Conv	0.23	0.22	0.22	0.23	0.22	0.22	0.22	0.27
SummaC-ZS	0.13	0.14	0.10	0.14	0.12	0.11	0.12	0.19
AlignScore	0.18	0.20	0.11	0.16	0.16	0.20	0.15	0.22
UniEval	0.68	0.71	0.67	0.65	0.72	0.32	0.65	0.70

Table 6: Mean factuality scores for each metric and perturbation type on SQuALITY.

average score differences are often close to zero due to cancellation effects, absolute changes capture the magnitude of metric instability regardless of direction. For a given domain, metric, and perturbation, we compute:

$$\Delta_{\text{abs}} = \frac{1}{N} \sum_{i=1}^{N} \left| M_{\text{pert}}^{(i)} - M_{\text{orig}}^{(i)} \right|, \qquad (5)$$

where $M_{\text{pert}}^{(i)}$ and $M_{\text{orig}}^{(i)}$ denote the factuality scores for the original and perturbed summaries of example i, respectively, and N is the number of summaries in the domain. This measure reflects the average magnitude of score variation induced by perturbations and serves as a robustness diagnostic.

In the legal domain (*LexAbSumm*), we observe the largest overall instability across both metrics and perturbations, as shown in Figure 8. Among the evaluated metrics, `AlignScore` exhibits the highest mean absolute score change, followed by `SummaC-ZS` and `UniEval`, indicating heightened sensitivity to surface-level changes in legally structured text. `MiniCheck` and `SummaC-Conv` show comparatively lower instability, while `BARTScore` exhibits the smallest absolute changes, consistent with its generally low baseline scores on this dataset. Across perturbations, *Negated* summaries produce by far the largest absolute score changes, followed by *Summarized* and *Paraphrased* variants. This pattern reflects the reliance of legal summaries on precise logical structure and domain-specific terminology, where even meaning-preserving changes can substantially alter cues used by factuality metrics.

For the narrative domain (*SQuALITY*), absolute score changes are smaller overall than in *LexAbSumm* but remain non-trivial, as shown in Figure 9. `UniEval`, `AlignScore`, and `MiniCheck` display the highest instability, while `SummaC-Conv` and `BARTScore` are comparatively stable. *Negated*

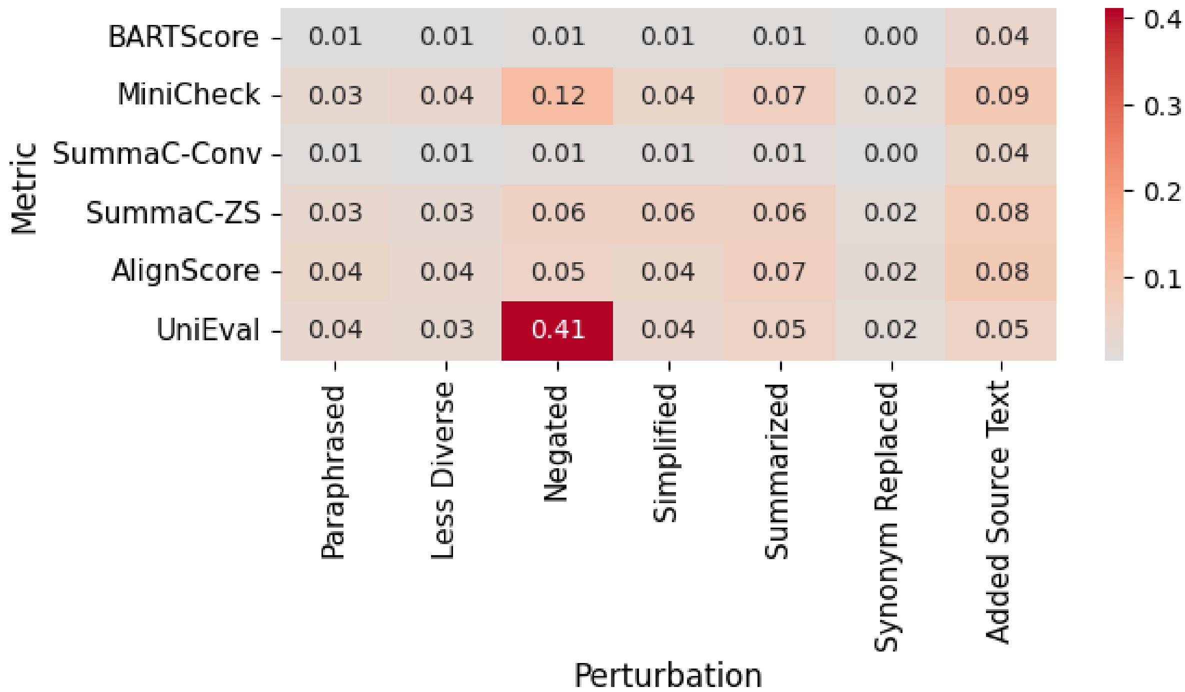

Figure 10: ScholarQABench: Metric score deltas under perturbations.

summaries have the highest score change, and perturbations that increase abstraction, particularly *Summarized* and *Added Source Text*, induce the largest score changes. This suggests that narrative summaries, which often compress temporal and causal structure, pose challenges for metrics when abstraction increases, even if factual content is preserved.

In the scientific multi-document domain (*ScholarQABench*), we observe the lowest absolute score changes across all metrics and perturbations, as illustrated in Figure 10. Although *UniEval* and *MiniCheck* still exhibit measurable sensitivity, the magnitude of instability is consistently lower than in the single-document domains. *Negated* and *Added Source Text* perturbations remain the most impactful, but their effects are attenuated. This relative stability likely arises from redundancy across multiple documents, where repeated evidence reduces the impact of localized reformulations on factuality assessment.

Overall, this analysis shows that factuality metric robustness varies substantially by domain, even under meaning-preserving perturbations. Legal text amplifies metric instability, narrative text exhibits moderate sensitivity to abstraction, and multi-document scientific text provides a stabilizing effect. These domain-specific patterns help explain the wide score distributions observed in Figure 2 and reinforce the need to evaluate factuality metrics under realistic long-document conditions.

F Code & Data Availability Statement

We release all perturbed summary data, along with the recipes used to generate it and the scripts required to reproduce our results, under the MIT License. The complete codebase and dataset artifacts will be made publicly available upon publication.

G Model Size & Budget

We describe all factuality metrics and the experimental setup in § 4. All experiments are conducted using a single NVIDIA H100 SXM5 80GB GPU.

H Software Package Parameters

- NLTK (Loper and Bird, 2002): We use the punkt sentence tokenizer for sentence tokenization.
- OpenAI GPT-4o: We use top p sampling at 50% with a temperature of 0 for all prompts.